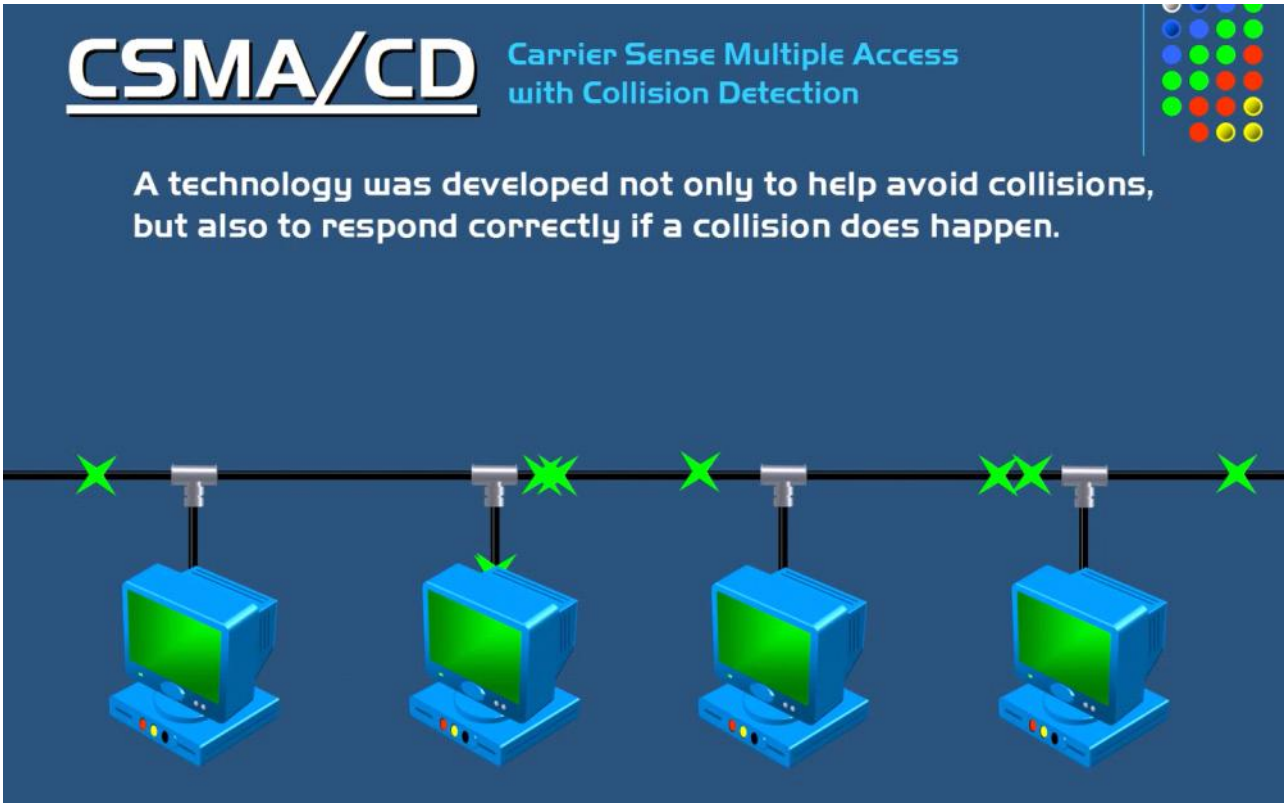
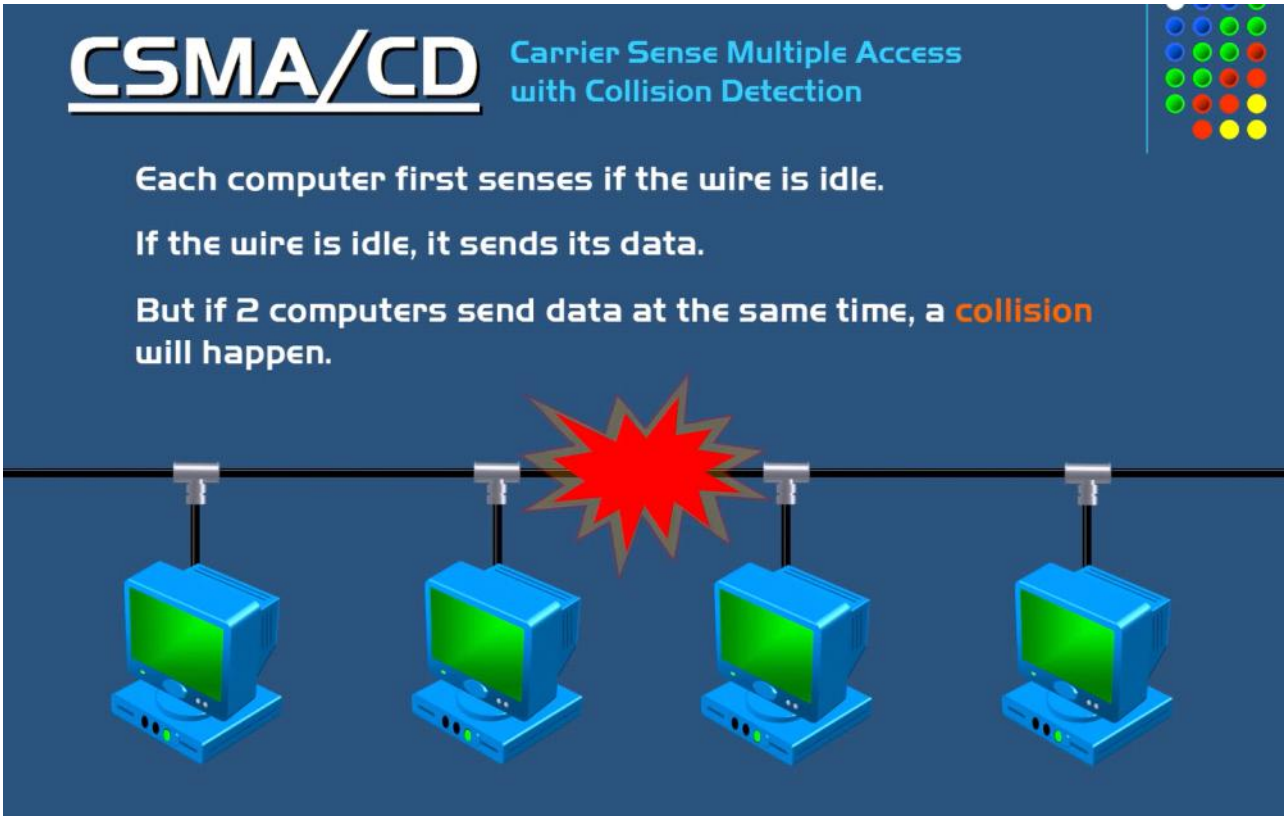
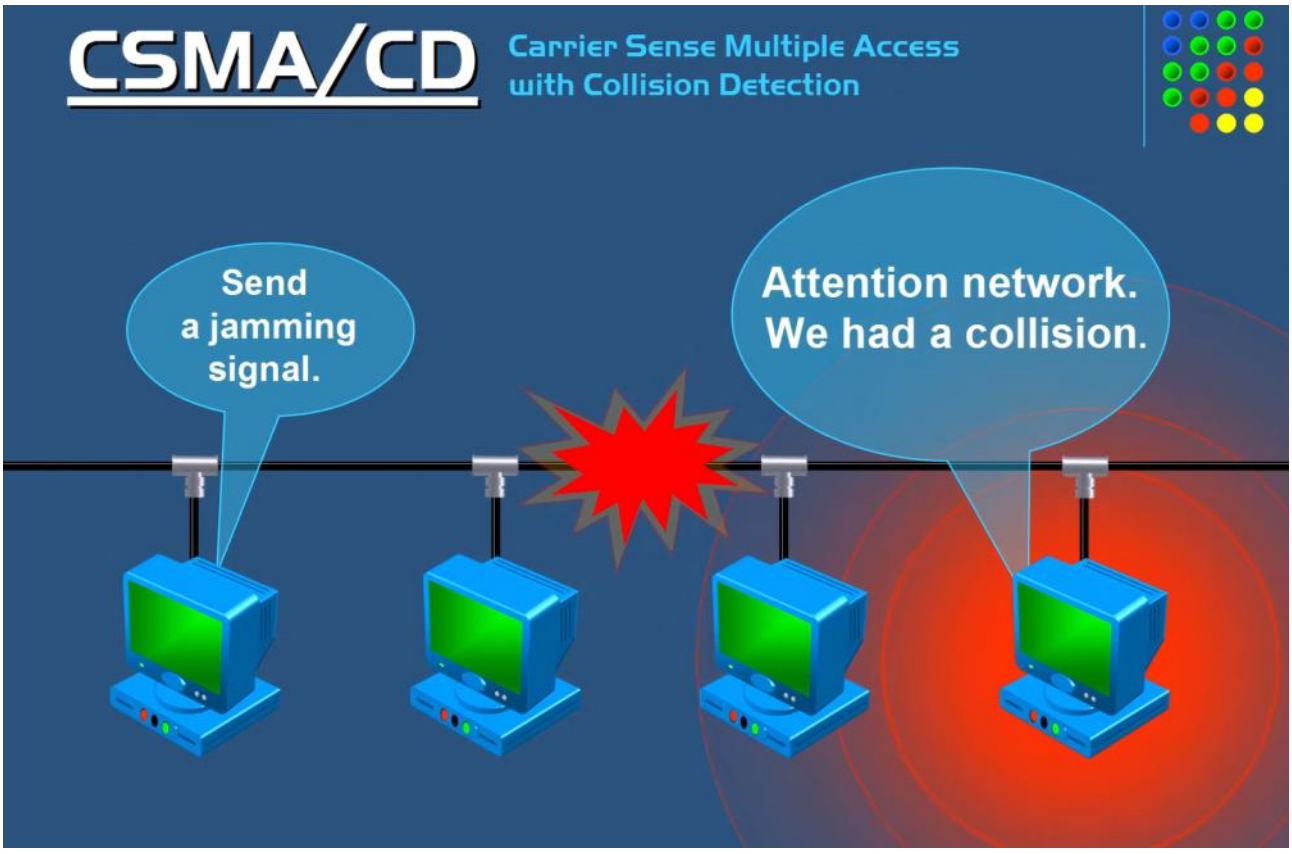
CSMA/CD & CSMA/CA

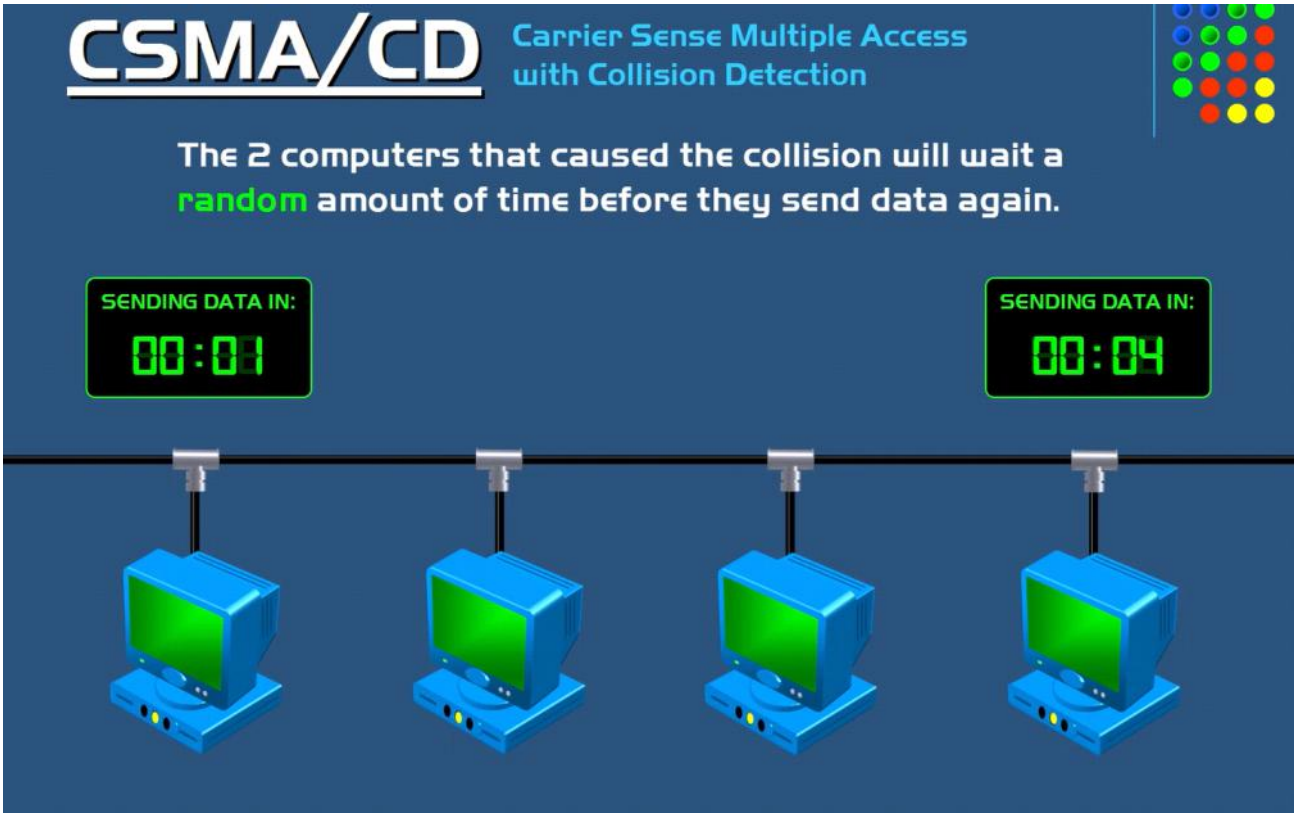
Saturday, April 11, 2026 12:30 PM

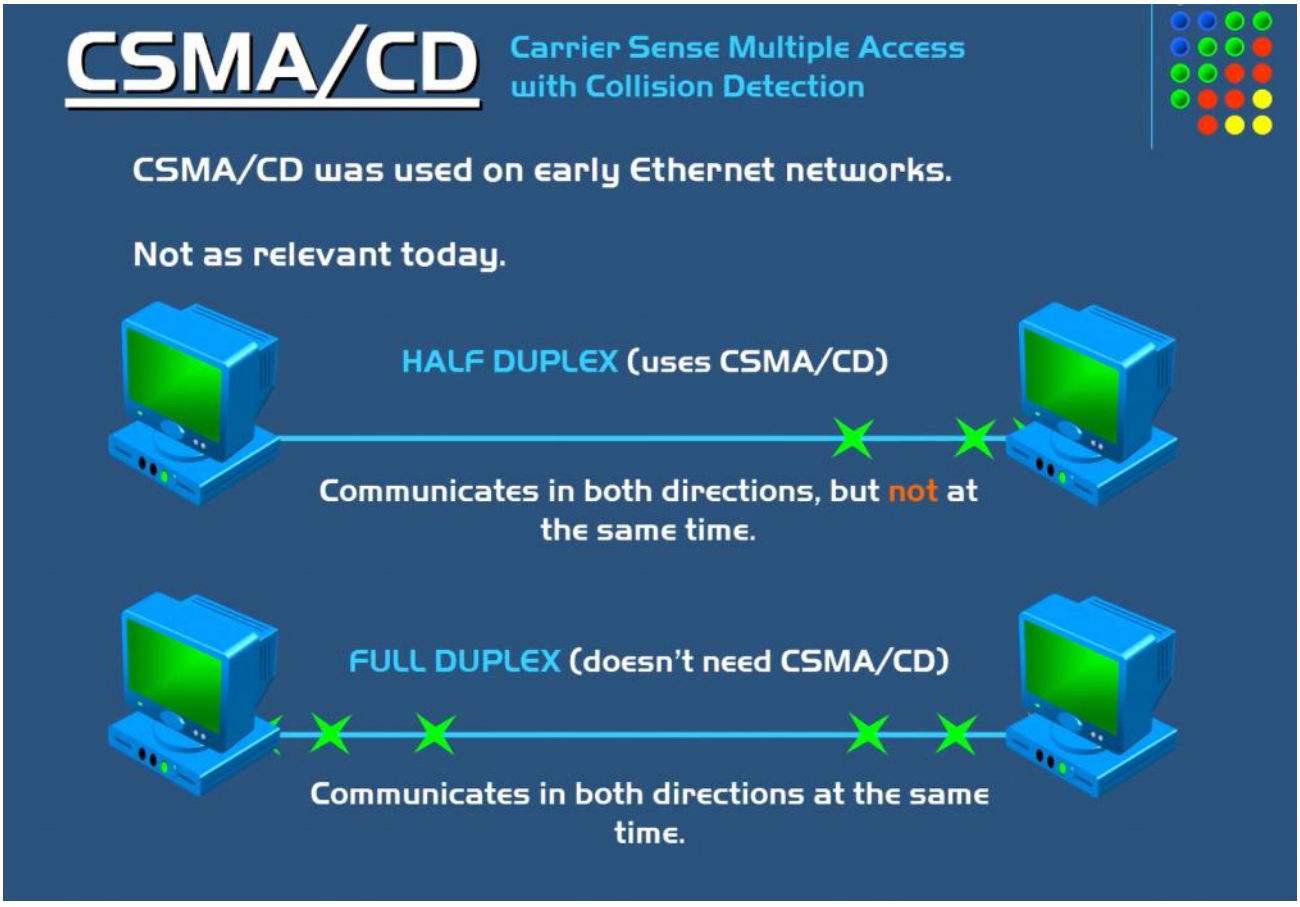
CSMA CD (used in old wired networks)





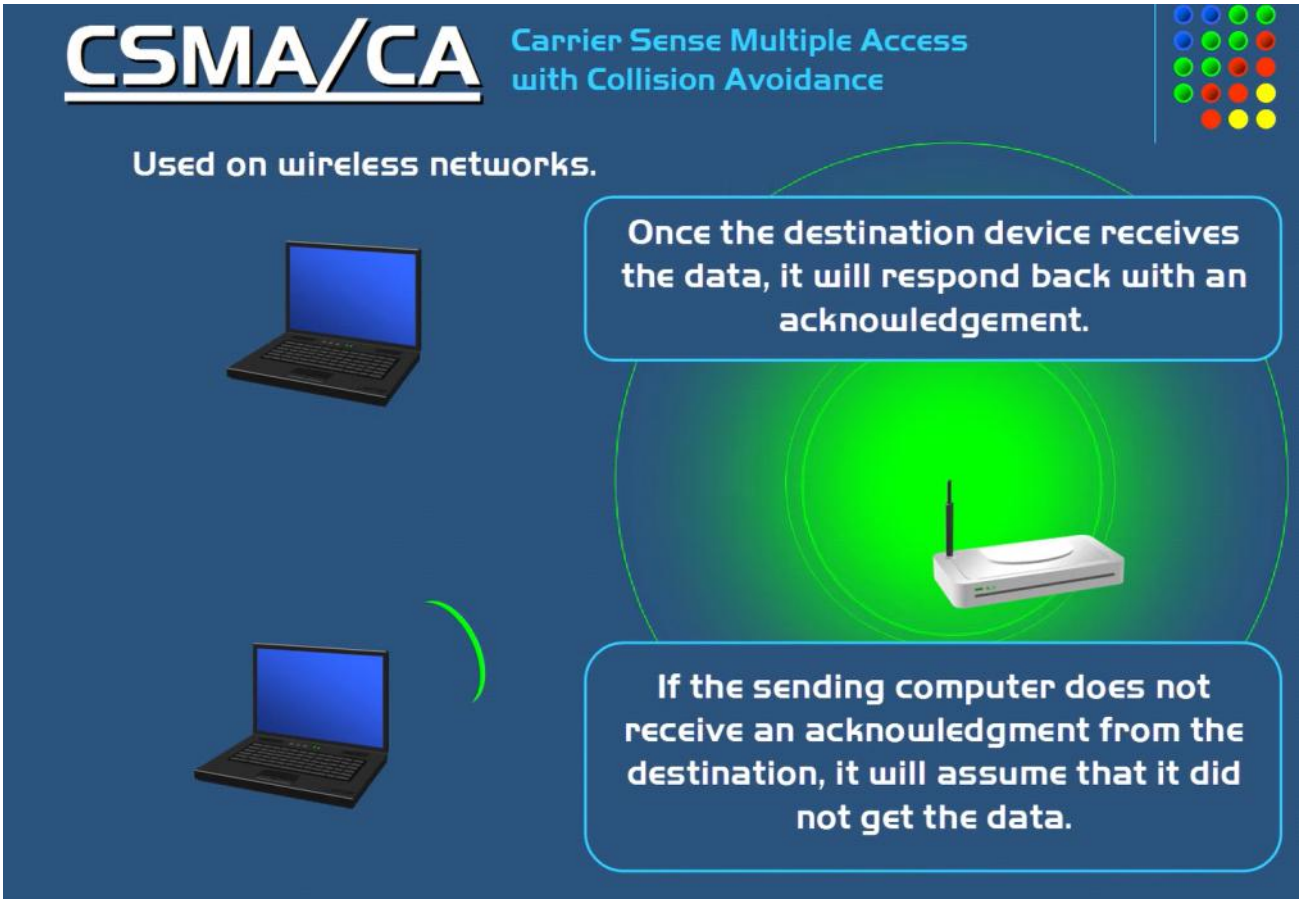

CSMA/CD
Carrier Sense Multiple Access
with Collision Detection
Send
a jamming
signal.
Attention network.
We had a collision.


CSMA/CD
Carrier Sense Multiple Access
with Collision Detection
The 2 computers that caused the collision will wait a
random amount of time before they send data again.
SENDING DATA IN:
00:01
SENDING DATA IN:
00:04



- Computers first check if the cable is free
- If free they send data
- If two send at same time collision happens
- Devices stop send a jamming signal and wait random time before retry
- Used in half duplex Ethernet old networks

CSMA CA (used in wireless networks)



- Devices try to avoid collisions instead of detecting
- They check channel wait random time then send
- Receiver sends acknowledgement after receiving data
- If no acknowledgement sender retries
- Optional RTS CTS helps control traffic by asking permission before sending

Key difference

- CSMA CD detects collisions after they happen
- CSMA CA tries to avoid collisions before they happen